TEACHER GUIDE

Cambridge IGCSE Business (0450 / 0264)

Production and capacity

4 Operations management
4.1 Production of goods and services: productivity



A revision lesson. Pupils run a car dealership in firms of four and revise production and capacity on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same.

1 Cars per staff

Round 1

The prep workshop.

Cars prepared per member of staff a day?

Cars prepared a week	**60 cars**
Staff	**6 staff**
Days worked a week	**5 days**

ANSWER

£2

60 ÷ 6 = 10 a week; ÷ 5 = 2 cars

Why: Productivity is output for each worker in a set time.

2 Raise output

Round 1

Your prep workshop is slow.

Which two would raise output per worker?

- **A** Buy better equipment
- **B** Raise your prices
- **C** Train staff in faster methods
- **D** Cut the hours staff work
- **E** Hire more staff with the same tools

ANSWER

✓ Buy better equipment
✓ Train staff in faster methods

Why: Productivity is output per worker; more staff raise output, not productivity.

3 Rank productivity

Round 2

Three workshops: cars prepared a week, and the staff who prepare them.

Order by productivity, highest first?

- **A** 24 cars, 4 staff
- **B** 45 cars, 15 staff
- **C** 40 cars, 8 staff

ANSWER

1. 24 cars, 4 staff
2. 40 cars, 8 staff
3. 45 cars, 15 staff

Why: Cars per worker: 6, 5 and 3. The busiest workshop is the least productive.

Topic tasks

4 New productivity

Round 2

More staff, more cars.

New cars prepared per member of staff?

Cars prepared a week	**28 cars**
Staff	**4 staff**
Extra staff	**3 staff**
Extra cars prepared a week	**28 cars**

ANSWER

8 cars

(28 + 28) ÷ (4 + 3) = 8 cars

Why: Productivity rose from 7 to 8 cars per worker.

5 One-off big order

Round 3

Your workshop is fully booked. A one-off big order arrives.

How should you handle it, and why?

A Pay overtime for this order

B Build a new prep bay

ANSWER

✔ Pay overtime for this order

Why: A one-off order does not justify the cost of a new bay.

6 Productivity rise %

Round 3

New tools and training.

Rise in cars per worker, as a %?

Cars a week, before	**16 cars**
Staff, before	**4 staff**
Cars a week, after	**18 cars**
Staff, after	**3 staff**

ANSWER

50%

16 ÷ 4 = 4; 18 ÷ 3 = 6; (6 − 4) ÷ 4 × 100 = 50%

Why: Each worker now prepares 6 cars a week, not 4.

Final questions

Each pupil answers 6 final questions on production and capacity. They come from these 10 questions.

1 Which situation is best suited to job production?
✔ **Restoring one classic car to a customer's wishes**

2 A parts maker produces twenty identical service kits, then resets the equipment to make a different kit. Which production method is this?
✔ **Batch production**

3 What is a likely advantage of flow production for standardised components?
✔ **High volumes at a low cost per unit**

4 A workshop queues every Saturday but has spare weekday capacity. What should it do before paying for a permanent extra bay?
✔ **First test moving work to weekdays**

5 A viral review causes a sudden rise in service bookings. What is the best short-term response?
✔ **Prioritise bookings and add trained help**

6 Labour productivity = total output ÷ number of employees. Example: 400 services ÷ 10 technicians = 40 per technician. Now 540 services and 12 technicians: what is labour productivity?
✔ **45 services per technician**

7 A workshop converts one van at a time to each customer's own design. Which production method is this?
✔ **Job production**

8 A parts maker produces thousands of identical brake pads on a continuous line. What is a main advantage of this flow production?
✔ **Low cost per unit from high volume**

9 A body shop paints each car in the exact colour and design the customer asks for. Which method of production is this?
✔ **Job production**

10 A workshop has 4 mechanics. Together they service 96 cars in a week. What is the labour productivity per mechanic?
✔ **24 cars a week**